

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.IT Semester V)
Artificial Intelligence Application Development and Jira

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PRACTICAL 7

AIM :- Machine Learning

6) Implement Decision Tree regression on a regression dataset.

CODE :-

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeRegressor

data = pd.read_csv("iris.csv")

X = data[['SepalLengthCm']]

y = data[['SepalWidthCm']]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)

model = DecisionTreeRegressor()

model.fit(X_train, y_train)

print("Training Accuracy : ",model.score(X_train, y_train)*100)

print("Testing Accuracy : ",model.score(X_test, y_test)*100)

new_data = pd.DataFrame([[11]],columns=['SepalLengthCm'])

prediction = model.predict(new_data)

print("Predicted SepalWidthCm : ",prediction[0])

print("Saqlain Khan T013")
```

OUTPUT :-

```
Training Accuracy : 31.638143919277752
Testing Accuracy : 14.145810934928061
Predicted SepalWidthCm : 3.0666666666666664
Saqlain Khan T013
```

7) Implement k-Nearest Neighbors (kNN) classifiers.

CODE :-

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

data = pd.read_csv("iris.csv")

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)

model = KNeighborsClassifier()

model.fit(X_train, y_train)

print("Training Accuracy : ",model.score(X_train, y_train)*100)

print("Testing Accuracy : ",model.score(X_test, y_test)*100)

prediction = model.predict([X_test[0]])

print("Predicted Class : ",prediction[0])

print("Saqlain Khan T013")
```

OUTPUT :-

```
Training Accuracy : 95.23809523809523
Testing Accuracy : 97.77777777777777
Predicted Class : 0
Saqlain Khan T013
```

8) Implement multiclass classification using Logistic Regression or Decision Tree.

CODE :-

```
from sklearn.datasets import load_iris
```

```

from sklearn.model_selection import train_test_split

from sklearn.tree import DecisionTreeClassifier

data = pd.read_csv("iris.csv")

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)

model = DecisionTreeClassifier()

model.fit(X_train, y_train)

print("Training Accuracy : ",model.score(X_train, y_train)*100)

print("Testing Accuracy : ",model.score(X_test, y_test)*100)

prediction = model.predict([X_test[0]])

print("Predicted Class : ",prediction[0])

print("Saqlain Khan T013")

```

OUTPUT :-

```

Training Accuracy : 100.0
Testing Accuracy : 95.55555555555556
Predicted Class : 0
Saqlain Khan T013

```

TASK

Perform Logistic Regression, Decision Tree, and kNN on the same dataset. Compare the Results.

Logistic Regression

CODE :-

```

import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LogisticRegression

data = pd.read_csv("netflix_titles.csv")

data['type'] = le.fit_transform(data['type'])

```

```

X = data[['release_year']]
y = data['type']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)
model = LogisticRegression()
model.fit(X_train, y_train)
print("Training Accuracy : ",model.score(X_train, y_train)*100)
print("Testing Accuracy : ",model.score(X_test, y_test)*100)
prediction = model.predict([X_test.iloc[0]])
print("Predicted Class : ",prediction[0])
print("Saqlain Khan T013")

```

OUTPUT :-

```

Training Accuracy : 70.37637897469176
Testing Accuracy : 67.83957623912221
Predicted Class : 0
Saqlain Khan T013

```

Decision Tree

CODE :-

```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
data = pd.read_csv("netflix_titles.csv")
data['type'] = le.fit_transform(data['type'])
X = data[['release_year']]
y = data['type']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)
model = DecisionTreeClassifier()
model.fit(X_train, y_train)
print("Training Accuracy : ",model.score(X_train, y_train)*100)
print("Testing Accuracy : ",model.score(X_test, y_test)*100)
prediction = model.predict([X_test.iloc[0]])
print("Predicted Class : ",prediction[0])

```

```
print("Saqlain Khan T013")
```

OUTPUT :-

```
Training Accuracy : 70.65217391304348
Testing Accuracy : 68.67196367763904
Predicted Class : 0
Saqlain Khan T013
```

kNN

CODE :-

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

data = pd.read_csv("netflix_titles.csv")
X = data[['release_year']]
y = data['type']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=1)

model = KNeighborsClassifier()

model.fit(X_train, y_train)

print("Training Accuracy : ",model.score(X_train, y_train)*100)
print("Testing Accuracy : ",model.score(X_test, y_test)*100)

prediction = model.predict([X_test.iloc[0]])

print("Predicted Class : ",prediction[0])

print("Saqlain Khan T013")
```

OUTPUT :-

```
Training Accuracy : 65.67164179104478
Testing Accuracy : 62.80741581536133
Predicted Class : 1
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```

Comparison

The results show that the **Decision Tree** performed best on the Netflix dataset, with a training accuracy of **70.65%** and testing accuracy of **68.67%**. Logistic Regression achieved **70.38% training accuracy** and **67.84% testing accuracy**, while kNN gave the lowest performance with **65.67% training accuracy** and **62.81% testing accuracy**.

In comparison, the Iris dataset gave much better results. Logistic Regression and kNN achieved **97.78% testing accuracy**, while Decision Tree achieved **95.56% testing accuracy**.

Therefore, the **Iris dataset performed significantly better** than the Netflix dataset. For the Netflix dataset, **Decision Tree was the best-performing algorithm**, whereas for the Iris dataset, **Logistic Regression and kNN performed better**.

Final conclusion

For the Netflix dataset:

- **Decision Tree** → Best testing accuracy: **68.67%**
- **Logistic Regression** → **67.84%**
- **kNN** → **62.81%**

For the Iris dataset:

- Logistic Regression → **97.78%**
- kNN → **97.78%**
- Decision Tree → **95.56%**

So, **Iris gives much higher accuracy than the Netflix dataset** using this setup. The Netflix model is only using **release_year** to predict whether something is a Movie or TV Show, so the classification is much harder with this single feature.